

TASK 1 — Docker Knowledge Assessment

Basic Docker

1. What is Docker?

Docker is a platform used to build, package, and run applications inside containers. It packages the application along with its dependencies, libraries, configuration, and required files into a portable unit. This helps the application run consistently across different environments such as a developer's laptop, testing server, and production server.

Docker also makes application deployment easier because containers can be created, started, stopped, removed, and recreated using Docker commands.

2. What is a container?

A container is a lightweight and isolated environment in which an application runs. It contains the application and the dependencies required by that application.

Containers are different from traditional virtual machines because they share the host operating system's kernel instead of requiring a complete guest operating system. Because of this, containers are generally faster to start and use fewer resources.

For example, a web application and a PostgreSQL database can run in separate containers while communicating with each other through a Docker network.

3. What is an image?

A Docker image is a read-only template used to create containers. It contains the application code, required dependencies, libraries, configuration files, and other components needed to run the application.

Images are built using a Dockerfile or can be downloaded from a container registry such as Docker Hub.

For example:

```
docker pull nginx
```

downloads the Nginx image, which can then be used to create an Nginx container.

4. What is Docker Hub?

Docker Hub is a cloud-based container registry used to store, share, and download Docker images. It provides many official and community-created images.

For example, developers can download images such as Nginx, PostgreSQL, Ubuntu, and Node.js from Docker Hub instead of creating every base image from scratch.

Docker images can also be pushed to Docker Hub so that other developers or deployment systems can use them.

5. What is the difference between an image and container?

A Docker image is a **template or blueprint**, while a container is an **instance created from that image**.

An image contains everything required to create and run an application. When Docker starts an image, it creates a container from that image.

For example, the nginx image can be used to create multiple Nginx containers. The image remains unchanged while each container has its own runtime environment.

In simple terms:

Docker Image

↓

Container

↓

Running Application

Dockerfile

1. What is Dockerfile?

A Dockerfile is a text file containing a series of instructions that Docker uses to build a custom image. It describes how the application environment should be created.

A Dockerfile can specify the base image, working directory, application files, dependencies, exposed ports, environment variables, and startup command.

For example:

```
FROM nginx:alpine
```

```
COPY index.html /usr/share/nginx/html/
```

```
EXPOSE 80
```

```
CMD ["nginx", "-g", "daemon off;"]
```

This Dockerfile creates an Nginx-based image and copies a web page into the Nginx document directory.

2. What does FROM do?

The FROM instruction specifies the base image used to build a Docker image. It is normally the first main instruction in a Dockerfile.

The base image provides an existing operating system environment or runtime that the application can use.

For example:

```
FROM python:3.12
```

uses Python 3.12 as the base environment.

Another example is:

```
FROM nginx:alpine
```

which uses an Alpine Linux-based Nginx image.

Using an appropriate base image can reduce the amount of configuration required when creating the application image.

3. What does COPY do?

The COPY instruction copies files or directories from the Docker build context into the Docker image.

It is commonly used to copy application source code, configuration files, HTML files, or dependency files into the image.

Example:

```
COPY app.py /app/app.py
```

This copies app.py from the build context into /app/app.py inside the image.

Another example:

```
COPY . /app
```

copies the project files into the /app directory, except files excluded by .dockerignore.

4. What does RUN do?

The RUN instruction executes a command while the Docker image is being built. It is commonly used to install application dependencies, packages, or tools.

For example:

```
RUN pip install -r requirements.txt
```

installs Python dependencies during the image build.

Another example:

```
RUN apt-get update
```

runs the package update command during the image-building process.

The result of a RUN instruction becomes part of the resulting Docker image.

5. What does CMD do?

The CMD instruction defines the default command that should run when a container starts.

For example:

```
CMD ["python", "app.py"]
```

tells Docker to start the Python application when the container runs.

Unlike RUN, which executes during image building, CMD is associated with the container's runtime.

A Dockerfile generally uses CMD to define how the application should start.

Compose

1. What is Docker Compose?

Docker Compose is a tool used to define and manage applications that require multiple Docker containers. The application's services, networks, volumes, ports, and environment variables can be configured in a Compose YAML file.

For example, an application may have:

Web Application

|

↓

PostgreSQL

Instead of manually creating and connecting each container, Docker Compose allows both services to be defined in one configuration.

A Compose project can then be started with:

`docker compose up`

and stopped with:

`docker compose down`

This makes multi-container application management easier and more consistent.

2. What is a service?

A service is a logical component of an application defined in the Docker Compose file. Each service normally represents a container or a group of containers created from the same configuration.

For example:

services:

web:

...

db:

...

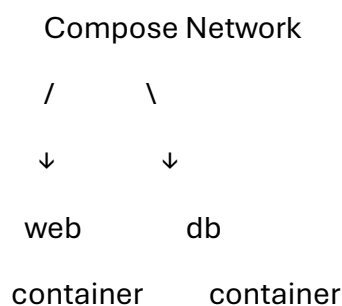
Here, web represents the web application and db represents the database.

Services can communicate with each other through the Compose network. For example, the web application can connect to PostgreSQL using the hostname db rather than localhost.

3. What is a Compose network?

A Compose network is a Docker network used by services belonging to the same Compose project. It allows containers to communicate with each other.

For example:



The web container can communicate with the database container through its Compose service name.

For example, if the database service is named db, the application can use:

Host: db

instead of:

Host: localhost

This is important because localhost inside a container refers to that same container, not another service.

4. Why use .env?

The .env file is commonly used to store configuration values and environment variables that may change between environments.

For example:

POSTGRES_USER=devops

POSTGRES_PASSWORD=password

POSTGRES_DB=appdb

These values can then be referenced from the Compose configuration.

Using `.env` makes configuration easier to manage and avoids putting environment-specific values directly throughout the Compose file.

Sensitive values such as passwords should also be handled carefully and should not accidentally be committed to a public GitHub repository.

5. Why use `.dockerignore`?

The `.dockerignore` file specifies files and directories that should not be included in the Docker build context.

For example:

```
.git
.gitignore
node_modules
__pycache__
.env
```

Ignoring unnecessary files can reduce the amount of data sent to the Docker daemon during the build. It can also make builds faster and help prevent unwanted files, temporary files, credentials, and development artifacts from being copied into the image.

Storage

1. What is a Docker volume?

A Docker volume is a storage mechanism managed by Docker that stores data outside the container's writable layer.

Volumes are commonly used for applications that need persistent data, especially databases such as PostgreSQL and MySQL.

For example:

PostgreSQL Container

|
↓

Docker Volume

|

↓

Persistent Database Data

If the PostgreSQL container is removed, the volume can remain available and be attached to a new PostgreSQL container.

2. Why do we need volumes?

Containers are designed to be replaceable. If important data is stored only inside a container and that container is removed, the data stored in its writable layer can be lost.

Volumes provide persistent storage that exists separately from the container.

For example, a PostgreSQL database can use a volume to store its database files. If the PostgreSQL container is recreated, the new container can use the existing volume and access the previous database data.

Volumes are therefore important for:

- Database storage
- Persistent application data
- Data that must survive container recreation
- Sharing persistent data between appropriate containers

3. What happens to container data when the container is removed?

Data stored only in the container's writable layer is normally removed along with the container. The container itself no longer exists, so its writable filesystem is also removed.

However, data stored in a separate Docker volume can remain even after the container is deleted.

For example:

Container

|

└ Temporary container data → Lost when container is removed

Volume

|

└ Database data → Remains after container removal

This is why databases should generally use persistent volumes when their data needs to survive container recreation.

Troubleshooting

1. How do you check container logs?

Container logs can be viewed using the `docker logs` command.

```
docker logs <container_name>
```

For example:

```
docker logs my-container
```

To continuously monitor new log messages, you can use:

```
docker logs -f my-container
```

For Docker Compose applications, logs can be checked using:

```
docker compose logs
```

Or for a specific service:

```
docker compose logs web
```

Logs are useful for identifying application errors, failed startup commands, database connection problems, configuration errors, and other runtime issues.

2. How do you find why a container stopped?

First, use:

```
docker ps -a
```

This displays both running and stopped containers. The `STATUS` column can show whether a container exited and may provide an exit code.

Next, check the container's logs:

```
docker logs <container_name>
```

Logs often provide the actual error that caused the application to stop.

You can also inspect the container:

```
docker inspect <container_name>
```

This provides detailed information about the container, including its configuration, state, networking, mounts, and exit information.

A typical troubleshooting process is:

Check container status

↓

```
docker ps -a
```

↓

Check logs

↓

```
docker logs <container>
```

↓

Inspect container

↓

```
docker inspect <container>
```

↓

Identify root cause

↓

Fix configuration/application

↓

Start and test again

This process helps determine whether the problem is caused by an application error, incorrect command, missing dependency, wrong port, database connection issue, environment variable, or another configuration problem.

TASK 2 — Create Final Project Repository

```

akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project$ mkdir -p app scripts docs
akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project$ ls
README.md app docs scripts
akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project$ touch Dockerfile docker-compose.yml .dockerignore .gitignore .env.example README.md
akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project$ ls
Dockerfile README.md app docker-compose.yml docs scripts
akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project$ ls -la
total 28
drwxr-xr-x 6 akhil_chary akhil_chary 4096 Aug 18 23:05 .
drwxr-xr-x 17 akhil_chary akhil_chary 4096 Aug 18 23:04 ..
-rw-r--r-- 1 akhil_chary akhil_chary 0 Aug 18 23:05 .dockerignore
-rw-r--r-- 1 akhil_chary akhil_chary 0 Aug 18 23:05 .env.example
drwxr-xr-x 8 akhil_chary akhil_chary 4096 Aug 18 23:04 .git
-rw-r--r-- 1 akhil_chary akhil_chary 0 Aug 18 23:05 .gitignore
-rw-r--r-- 1 akhil_chary akhil_chary 0 Aug 18 23:05 Dockerfile
-rw-r--r-- 1 akhil_chary akhil_chary 72 Aug 18 23:05 README.md
drwxr-xr-x 2 akhil_chary akhil_chary 4096 Aug 18 23:05 app
-rw-r--r-- 1 akhil_chary akhil_chary 0 Aug 18 23:05 docker-compose.yml
drwxr-xr-x 2 akhil_chary akhil_chary 4096 Aug 18 23:05 docs
drwxr-xr-x 2 akhil_chary akhil_chary 4096 Aug 18 23:05 scripts
akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project$ nano .gitignore
akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project$ nano .dockerignore
akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project$ git status
On branch main
Your branch is up to date with 'origin/main'.

```

Activate Windows
Go to Settings to activate Windows.

```

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .dockerignore
        .env.example
        .gitignore
        Dockerfile
        docker-compose.yml

nothing added to commit but untracked files present (use "git add" to track)
akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project$ git add .
akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project$ git ststus
git: 'ststus' is not a git command. See 'git --help'.

```

```

The most similar command is
status
akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project$ git status
On branch main
Your branch is up to date with 'origin/main'.

```

```

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   .dockerignore
        new file:   .env.example
        new file:   .gitignore
        new file:   Dockerfile
        new file:   docker-compose.yml

```

```

akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project$ git commit -m "Initialize Docker final project structure"
[main 803f1f3] Initialize Docker final project structure

```

Activate Windows
Go to Settings to activate Windows.

```

akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project$ git commit -m "Initialize Docker final project structure"
[main 803f1f3] Initialize Docker final project structure
5 files changed, 11 insertions(+)
create mode 100644 .dockerignore
create mode 100644 .env.example
create mode 100644 .gitignore
create mode 100644 Dockerfile
create mode 100644 docker-compose.yml
akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project$ git push -u origin main
Username for 'https://github.com': akhilchary81-lab
Password for 'https://akhilchary81-lab@github.com':
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 495 bytes | 495.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/akhilchary81-lab/devops-docker-final-project.git
   f101ee4..803f1f3  main -> main
branch 'main' set up to track 'origin/main'.

```

Activate Windows

TASK 3 — Build a Small Application

```
akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project$ pwd
/home/akhil_chary/devops-docker-final-project
akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project$ cd app
akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project/app$ touch app.py
akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project/app$ ls
app.py
akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project/app$ python3 -m venv venv
The virtual environment was not created successfully because ensurepip is not
available. On Debian/Ubuntu systems, you need to install the python3-venv
package using the following command.

    apt install python3.12-venv

You may need to use sudo with that command. After installing the python3-venv
package, recreate your virtual environment.

Failing command: /home/akhil_chary/devops-docker-final-project/app/venv/bin/python3
```

```
akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project/app$ sudo apt update
[sudo] password for akhil_chary:
Sorry, try again.
[sudo] password for akhil_chary:
Get:1 https://download.docker.com/linux/ubuntu noble InRelease [48.5 kB]
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:3 http://archive.ubuntu.com/ubuntu noble InRelease
Get:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:5 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages [64.1 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [952 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1216 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [204 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [285 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [46.4 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1201 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [181 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1683 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [336 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [388 kB]
Get:17 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [240 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1463 kB]
```

```

Setting up python3-pip-whl (24.0+dfsg-1ubuntu1.3) ...
Setting up python3.12-venv (3.12.3-1ubuntu0.15) ...
akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project/app$ python3 -m venv venv
akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project/app$ source venv/bin/activate
(venv) akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project/app$ pip install flask
Collecting flask
  Downloading flask-3.1.3-py3-none-any.whl.metadata (3.2 kB)
Collecting blinker>=1.9.0 (from flask)
  Downloading blinker-1.9.0-py3-none-any.whl.metadata (1.6 kB)
Collecting click>=8.1.3 (from flask)
  Downloading click-8.4.2-py3-none-any.whl.metadata (2.6 kB)
Collecting itsdangerous>=2.2.0 (from flask)
  Downloading itsdangerous-2.2.0-py3-none-any.whl.metadata (1.9 kB)
Collecting jinja2>=3.1.2 (from flask)
  Downloading jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting markupsafe>=2.1.1 (from flask)
  Downloading markupsafe-3.0.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (2.7 kB)
Collecting werkzeug>=3.1.0 (from flask)
  Downloading werkzeug-3.1.8-py3-none-any.whl.metadata (4.0 kB)
Downloading flask-3.1.3-py3-none-any.whl (103 kB)
   103.4/103.4 kB 812.3 kB/s eta 0:00:00
Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
   119.2/119.2 kB 2.3 MB/s eta 0:00:00
Downloading click-8.4.2-py3-none-any.whl (119 kB)
   134.9/134.9 kB 1.7 MB/s eta 0:00:00
Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
   134.9/134.9 kB 1.7 MB/s eta 0:00:00
Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
   134.9/134.9 kB 1.7 MB/s eta 0:00:00
Downloading markupsafe-3.0.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (22 kB)

```

```

^C(venv) akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project/app$ python app.py
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.20.157.14:5000
Press CTRL+C to quit
172.20.144.1 - - [19/Aug/2026 00:17:18] "GET / HTTP/1.1" 200 -
172.20.144.1 - - [19/Aug/2026 00:17:19] "GET /favicon.ico HTTP/1.1" 404 -
127.0.0.1 - - [19/Aug/2026 00:17:41] "GET /health HTTP/1.1" 200 -
127.0.0.1 - - [19/Aug/2026 00:17:41] "GET /favicon.ico HTTP/1.1" 404 -
^C(venv) akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project/app$ cd ..
(venv) akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project$ nano .dockerignore
(venv) akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project$ cd app
(venv) akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project/app$ source venv/bin/activate
(venv) akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project/app$ python app.py
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.20.157.14:5000
Press CTRL+C to quit
127.0.0.1 - - [19/Aug/2026 00:20:55] "GET / HTTP/1.1" 200 -
^C(venv) akhil_chary@DESKTOP-1EMNQEB:~/devops-docker-final-project/app$

```

← → ↻ ⚠ Not secure 172.20.157.14:5000

DevOps Training Application

Application Status: Running

Environment: Development

Database: Connected

← → ↻ ⓘ localhost:5000/health

Pretty-print ☐

```
{"application": "running", "status": "healthy"}
```

TASK 4 — Create Production-Style Dockerfile

```
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ cd app
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project/app$ touch app/app.py
touch: cannot touch 'app/app.py': No such file or directory
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project/app$ touch app.py
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project/app$ ls -la
total 0
drwxrwxrwx 1 akhil_chary akhil_chary 4096 Aug 19 01:37
drwxrwxrwx 1 akhil_chary akhil_chary 4096 Aug 19 01:43
-rwxrwxrwx 1 akhil_chary akhil_chary 0 Aug 19 01:22 .gitkeep
-rwxrwxrwx 1 akhil_chary akhil_chary 0 Aug 19 01:44 app.py
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project/app$ touch requirements.txt
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project/app$ ls -la
total 0
drwxrwxrwx 1 akhil_chary akhil_chary 4096 Aug 19 01:45
drwxrwxrwx 1 akhil_chary akhil_chary 4096 Aug 19 01:43
-rwxrwxrwx 1 akhil_chary akhil_chary 0 Aug 19 01:22 .gitkeep
-rwxrwxrwx 1 akhil_chary akhil_chary 0 Aug 19 01:44 app.py
-rwxrwxrwx 1 akhil_chary akhil_chary 0 Aug 19 01:45 requirements.txt
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project/app$ echo "Flask" > requirements.txt
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project/app$ cat requirements.txt
Flask
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project/app$ nano app.py
```

```

akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project/app$ ls -la
total 0
drwxrwxrwx 1 akhil_chary akhil_chary 4096 Aug 19 01:46 .
drwxrwxrwx 1 akhil_chary akhil_chary 4096 Aug 19 01:43 ..
-rwxrwxrwx 1 akhil_chary akhil_chary 0 Aug 19 01:22 .gitkeep
-rwxrwxrwx 1 akhil_chary akhil_chary 508 Aug 19 01:46 app.py
-rwxrwxrwx 1 akhil_chary akhil_chary 6 Aug 19 01:45 requirements.txt
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project/app$ cd ..
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ pwd
/mnt/c/WINDOWS/system32/devops-docker-final-project
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ ls -la
total 0
drwxrwxrwx 1 akhil_chary akhil_chary 4096 Aug 19 01:43 .
drwxrwxrwx 1 akhil_chary akhil_chary 4096 Aug 19 01:21 ..
-rwxrwxrwx 1 akhil_chary akhil_chary 54 Aug 19 01:21 .dockerignore
-rwxrwxrwx 1 akhil_chary akhil_chary 0 Aug 19 01:21 .env.example
drwxrwxrwx 1 akhil_chary akhil_chary 4096 Aug 19 01:23 .git
-rwxrwxrwx 1 akhil_chary akhil_chary 30 Aug 19 01:21 .gitignore
-rwxrwxrwx 1 akhil_chary akhil_chary 0 Aug 19 01:21 Dockerfile
-rwxrwxrwx 1 akhil_chary akhil_chary 72 Aug 19 01:21 README.md
drwxrwxrwx 1 akhil_chary akhil_chary 4096 Aug 19 01:46 app
-rwxrwxrwx 1 akhil_chary akhil_chary 0 Aug 19 01:21 docker-compose.yml
drwxrwxrwx 1 akhil_chary akhil_chary 4096 Aug 19 01:22 docs
drwxrwxrwx 1 akhil_chary akhil_chary 4096 Aug 19 01:22 scripts
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ nano Dockerfile
Activate Windows

```

```

akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker build -t devops-final:v1 .
[+] Building 30.2s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 350B
=> [internal] load metadata for docker.io/library/python:3.12-slim
=> [internal] load .dockerignore
=> => transferring context: 94B
=> [internal] load build context
=> => transferring context: 662B
=> [1/6] FROM docker.io/library/python:3.12-slim@sha256:2c941e860699f878900b0edc2403613c234d4b32eda3cc9fa7036991a
=> => resolve docker.io/library/python:3.12-slim@sha256:2c941e860699f878900b0edc2403613c234d4b32eda3cc9fa7036991a2
=> => sha256:6760bfe2ff00c4530bc73b2f88a1e9615a56c9a77028f41f8bb4b978d08b8439 248B / 248B
=> => sha256:b952fe9f6810de5dac5d24a3983aee5f3c3884f7359a533ba325e48ecb745994 12.12MB / 12.12MB
=> => sha256:fa18dfb1257a9c1afc75e233c55b0195dd02b8d6d18dd7e24c10238b039e7742 1.29MB / 1.29MB
=> => extracting sha256:fa18dfb1257a9c1afc75e233c55b0195dd02b8d6d18dd7e24c10238b039e7742
=> => extracting sha256:b952fe9f6810de5dac5d24a3983aee5f3c3884f7359a533ba325e48ecb745994
=> => extracting sha256:6760bfe2ff00c4530bc73b2f88a1e9615a56c9a77028f41f8bb4b978d08b8439
=> [2/6] WORKDIR /app
=> [3/6] COPY app/requirements.txt .
=> [4/6] RUN pip install --no-cache-dir -r requirements.txt
=> [5/6] COPY app/ .
=> [6/6] RUN useradd --create-home appuser && chown -R appuser:appuser /app
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:d771cbf787c4238066130d24e4ee6a46db1407a07683612bc1224a19f43d4b7e
=> => exporting config sha256:91556055fa0e42a2a025d5cc0b5fe00241a13db79a3010f46499498bde3e7bad
=> => exporting attestation manifest sha256:eb09041e7e397f442382f49166357a20ecdc463c818e824030e15ae08f2666d7
=> => exporting manifest list sha256:fef04589dbb2d7104f79963fc6fcb5966bb2f6f08e21f03220b499d0da4c7e68
=> => naming to docker.io/library/devops-final:v1

```

```

=> [3/6] COPY app/requirements.txt .
=> [4/6] RUN pip install --no-cache-dir -r requirements.txt
=> [5/6] COPY app/ .
=> [6/6] RUN useradd --create-home appuser && chown -R appuser:appuser /app
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:d771cbf787c4238066130d24e4ee6a46db1407a07683612bc1224a19f43d4b7e
=> => exporting config sha256:91556055fa0e42a2a025d5cc0b5fe00241a13db79a3010f46499498bde3e7bad
=> => exporting attestation manifest sha256:eb09041e7e397f442382f49166357a20ecdc463c818e824030e15ae08f2666d7
=> => exporting manifest list sha256:fef04589dbb2d7104f79963fc6fcb5966bb2f6f08e21f03220b499d0da4c7e68
=> => naming to docker.io/library/devops-final:v1
=> => unpacking to docker.io/library/devops-final:v1
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
devops-final:v1	fef04589dbb2	184MB	44.8MB	
docker-compose-day9-web:latest	0a885c785f9c	92.7MB	26.1MB	U
nginx:alpine	4a73073bd557	93.6MB	27MB	U
postgres:16-alpine	cf78e76683b9	420MB	117MB	U
troubleshooting-app:v2	19e49d2d648e	92.7MB	26.1MB	U

akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project\$

TASK 5 — Create Docker Compose Architecture

```
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ nano docker-compose.yml
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose config
name: devops-docker-final-project
services:
  app:
    build:
      context: /mnt/c/WINDOWS/system32/devops-docker-final-project
      dockerfile: Dockerfile
    container_name: devops-app
    depends_on:
      database:
        condition: service_healthy
        required: true
    environment:
      APP_ENV: development
      DB_HOST: database
      DB_NAME: devopsdb
      DB_PASSWORD: devops_password
      DB_PORT: "5432"
      DB_USER: devops
    image: devops-final:v1
    networks:
      devops-network: null
    ports:
      - mode: ingress
        target: 5000
        published: "5000"
        protocol: tcp
```

Activate Windows
Go to Settings to activate Windows.

```
Restart: unless-stopped
database:
  container_name: devops-database
  environment:
    POSTGRES_DB: devopsdb
    POSTGRES_PASSWORD: devops_password
    POSTGRES_USER: devops
  healthcheck:
    test:
      - CMD-SHELL
      - pg_isready -U devops -d devopsdb
    timeout: 5s
    interval: 10s
    retries: 5
  image: postgres:16-alpine
  networks:
    devops-network: null
  ports:
    - mode: ingress
      target: 5432
      published: "5432"
      protocol: tcp
  restart: unless-stopped
  volumes:
    - type: volume
      source: postgres-data
      target: /var/lib/postgresql/data
      volume: {}
```

Activate Windows
Go to Settings to activate Windows.

```

    published: "5432"
    protocol: tcp
    restart: unless-stopped
    volumes:
      - type: volume
        source: postgres-data
        target: /var/lib/postgresql/data
        volume: {}
  networks:
    devops-network:
      name: devops-docker-final-project_devops-network
      driver: bridge
  volumes:
    postgres-data:
      name: devops-docker-final-project_postgres-data
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose up -d
[+] up 4/4
  Network devops-docker-final-project_devops-network Created 0.1s
  Volume devops-docker-final-project_postgres-data Created 0.0s
  Container devops-database Healthy 11.6s
  Container devops-app Started 11.9s
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose ps

```

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS	PORT
devops-app	devops-final:v1	"python app.py"	app	26 seconds ago	Up 12 seconds	0.0.
devops-database	postgres:16-alpine	"docker-entrypoint.s..."	database	27 seconds ago	Up 25 seconds (healthy)	0.0.

Go to Settings to activate Windows.

```

akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$

```

← → ↻ ⓘ localhost:5000

DevOps Training Application

Application Status: Running

Environment: Development

Database: Connected

← → ↻ ⓘ localhost:5000/health

Pretty-print ☐

```
{"application": "running", "status": "healthy"}
```

TASK 6 — Environment & Secret Handling

```

akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ touch .env
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ nano .env
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ nano .env.example
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ nano .gitignore
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ ls -la
total 4
drwxrwxrwx 1 akhil_chary akhil_chary 4096 Aug 19 02:07 .
drwxrwxrwx 1 akhil_chary akhil_chary 4096 Aug 19 01:21 ..
-rwxrwxrwx 1 akhil_chary akhil_chary  54 Aug 19 01:21 .dockerignore
-rwxrwxrwx 1 akhil_chary akhil_chary 120 Aug 19 02:05 .env
-rwxrwxrwx 1 akhil_chary akhil_chary  79 Aug 19 02:06 .env.example
drwxrwxrwx 1 akhil_chary akhil_chary 4096 Aug 19 01:23 .git
-rwxrwxrwx 1 akhil_chary akhil_chary  30 Aug 19 02:07 .gitignore
-rwxrwxrwx 1 akhil_chary akhil_chary 311 Aug 19 01:47 Dockerfile
-rwxrwxrwx 1 akhil_chary akhil_chary  72 Aug 19 01:21 README.md
drwxrwxrwx 1 akhil_chary akhil_chary 4096 Aug 19 01:46 app
-rwxrwxrwx 1 akhil_chary akhil_chary 1022 Aug 19 01:55 docker-compose.yml
drwxrwxrwx 1 akhil_chary akhil_chary 4096 Aug 19 01:22 docs
drwxrwxrwx 1 akhil_chary akhil_chary 4096 Aug 19 01:22 scripts

```

```

akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   .env.example
        modified:   .gitignore
        modified:   Dockerfile
        modified:   docker-compose.yml

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        app/app.py
        app/requirements.txt

no changes added to commit (use "git add" and/or "git commit -a")
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git add .env.example .gitignore
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git commit -m "Add environment configuration template"
[main ab1045f] Add environment configuration template
 2 files changed, 6 insertions(+), 1 deletion(-)

```

```

akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git push origin main
Username for 'https://github.com': akhilchary81-lab
Password for 'https://akhilchary81-lab@github.com':
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 397 bytes | 16.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0
To https://github.com/akhilchary81-lab/devops-docker-final-project.git
 929300c..ab1045f  main -> main

```

Activate Windows
Go to Settings to activate Windows.

TASK 7 — Run the Complete Application

```

akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose build
[+] Building 3.8s (13/13) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 566B 0.0s
=> [internal] load build definition from Dockerfile 0.1s
=> => transferring dockerfile: 350B 0.0s
=> [internal] load metadata for docker.io/library/python:3.12-slim 1.8s
=> [internal] load .dockerignore 0.1s
=> => transferring context: 94B 0.0s
=> [1/6] FROM docker.io/library/python:3.12-slim@sha256:2c941e860699f878900b0edc2403613c234d4b32eda3cc9fa7036991a2 0.1s
=> => resolve docker.io/library/python:3.12-slim@sha256:2c941e860699f878900b0edc2403613c234d4b32eda3cc9fa7036991a2 0.1s
=> [internal] load build context 0.1s
=> => transferring context: 123B 0.1s
=> CACHED [2/6] WORKDIR /app 0.0s
=> CACHED [3/6] COPY app/requirements.txt . 0.0s
=> CACHED [4/6] RUN pip install --no-cache-dir -r requirements.txt 0.0s
=> CACHED [5/6] COPY app/ . 0.0s
=> CACHED [6/6] RUN useradd --create-home appuser && chown -R appuser:appuser /app 0.0s
=> exporting to image 0.3s
=> => exporting layers 0.0s
=> => exporting manifest sha256:5d4a6dc141d5036593b3805b9510b7cf948403f5a8a6a3e39c3169d30aeee3d4 0.0s
=> => exporting config sha256:455e54dab5fe7e95b398bdb1c157ab9c9cd6e137b20bd5872ad958758c741b 0.0s
=> => exporting attestation manifest sha256:f8032bda29652ec565ad0eb62be20b7e8622feff919ff9aa0c23ec8f28ca788d 0.0s
=> => exporting manifest list sha256:87f5afd1c4f4733868c95306ee518c36fab4ae678e5d4080371cb555bee1baf5 0.0s
=> => naming to docker.io/library/devops-final:v1 0.0s
=> => unpacking to docker.io/library/devops-final:v1 0.0s
=> resolving provenance for metadata file 0.0s
[+] build 1/1
Image devops-final:v1 Built 4.0s

```

```

akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
devops-final:v1	87f5afd1c4f4	184MB	44.8MB	
docker-compose-day9-web:latest	0a885c785f9c	92.7MB	26.1MB	U
nginx:alpine	4a73073bd557	93.6MB	27MB	U
postgres:16-alpine	cf78e76683b9	420MB	117MB	U
troubleshooting-app:v2	19e49d2d648e	92.7MB	26.1MB	U

```

akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose up -d
[+] up 2/2
  Container devops-database Healthy 11.2s
  Container devops-app Started 11.8s
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose ps

```

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS	PORT
devops-app	devops-final:v1	"python app.py"	app	36 seconds ago	Up 24 seconds	0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
devops-database	postgres:16-alpine	"docker-entrypoint.s..."	database	23 minutes ago	Up 23 minutes (healthy)	0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp

```

akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker ps

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
f98f68809be2	devops-final:v1	"python app.py"	55 seconds ago	Up 43 seconds	0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
04a06a87e853	postgres:16-alpine	"docker-entrypoint.s..."	23 minutes ago	Up 23 minutes (healthy)	0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
73acb58f639e	docker-compose-day9-web	"/docker-entrypoint..."	23 hours ago	Up 3 hours (healthy)	0.0.0.0:9090->80/tcp, [::]:9090->80/tcp
f5953c4125a7	postgres:16-alpine	"docker-entrypoint.s..."	23 hours ago	Up 3 hours (healthy)	5432/tcp

```

akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose logs app
devops-app | * Serving Flask app 'app'
devops-app | * Debug mode: off
devops-app | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
devops-app | * Running on all addresses (0.0.0.0)
devops-app | * Running on http://127.0.0.1:5000
devops-app | * Running on http://172.21.0.3:5000
devops-app | Press CTRL+C to quit
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose logs -f app
devops-app | * Serving Flask app 'app'
devops-app | * Debug mode: off
devops-app | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
devops-app | * Running on all addresses (0.0.0.0)
devops-app | * Running on http://127.0.0.1:5000
devops-app | * Running on http://172.21.0.3:5000
devops-app | Press CTRL+C to quit
^C
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose logs database
devops-database | The files belonging to this database system will be owned by user "postgres".
devops-database | This user must also own the server process.
devops-database |
devops-database | The database cluster will be initialized with locale "en_US.utf8".
devops-database | The default database encoding has accordingly been set to "UTF8".
devops-database | The default text search configuration will be set to "english".
devops-database |
devops-database | Data page checksums are disabled.

```

```

devops-database | PostgreSQL init process complete; ready for start up.
devops-database |
devops-database | 2026-08-19 08:56:52.556 UTC [1] LOG: starting PostgreSQL 16.15 on x86_64-pc-linux-musl, compiled by gcc (Alpine 15.2.0) 15.2.0, 64-bit
devops-database | 2026-08-19 08:56:52.557 UTC [1] LOG: listening on IPv4 address "0.0.0.0", port 5432
devops-database | 2026-08-19 08:56:52.557 UTC [1] LOG: listening on IPv6 address ":::", port 5432
devops-database | 2026-08-19 08:56:52.576 UTC [1] LOG: listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
devops-database | 2026-08-19 08:56:52.592 UTC [57] LOG: database system was shut down at 2026-08-19 08:56:52 UTC
devops-database | 2026-08-19 08:56:52.602 UTC [1] LOG: database system is ready to accept connections
devops-database | 2026-08-19 09:02:17.523 UTC [55] LOG: checkpoint starting: time
devops-database | 2026-08-19 09:02:20.524 UTC [55] LOG: checkpoint complete: wrote 31 buffers (0.2%); 0 WAL file(s) added, 0 removed, 0 recycled; write=2.816 s, sync=0.087 s, total=3.001 s; sync files=11, longest=0.021 s, average=0.008 s; distance=139 kB, estimate=139 kB; lsn=0/1948D90, redo lsn=0/1948D58
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ curl -i http://localhost:5000/health
HTTP/1.1 200 OK
Server: Werkzeug/3.1.8 Python/3.12.14
Date: Wed, 19 Aug 2026 09:24:17 GMT
Content-Type: application/json
Content-Length: 45
Connection: close

{"application": "running", "status": "healthy"}

```

```

akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose ps
NAME                IMAGE                COMMAND                SERVICE    CREATED        STATUS                PORTS
devops-app           devops-final:v1      "python app.py"        app         5 minutes ago  Up 4 minutes         0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
devops-database      postgres:16-alpine   "docker-entrypoint.s..." database    27 minutes ago  Up 27 minutes (healthy) 0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker inspect --format='{{.State.Health.Status}}' devops-database
healthy
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose exec database pg_isready -U devops -d devopsdb
/var/run/postgresql:5432 - accepting connections
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
eb0a7452964a        bridge             bridge             local
94b570376518        devops-docker-final-project_devops-network    bridge             local
5777044da3bd        docker-compose-day8_default    bridge             local
7aa49dac030d        docker-compose-day9_app-network    bridge             local
4129d5dc9746        host               host               local
41c3124c4dc7        none               null               local
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker network inspect devops-docker-final-project_devops-network
[
  {
    "Name": "devops-docker-final-project_devops-network",
    "Id": "94b570376518f9f7dcf8dce757769f312b0a738e9db6b47833c6f9aeed8d7260",
    "Created": "2026-08-19T01:56:43.651133876-07:00",
    "Scope": "local",

```

```
"com.docker.compose.version": "5.4.0"
},
"Containers": {
  "04a06a87e85313ebac6142a9c1d682093cd48fd36421f9bba240f2dbeff34e02": {
    "Name": "devops-database",
    "EndpointID": "246cd24be88e58b15a2da9b425893b774abc1484f591aa1e2ecbf239c4412386",
    "MacAddress": "ee:ae:be:f6:52:62",
    "IPv4Address": "172.21.0.2/16",
    "IPv6Address": ""
  },
  "f98f68809be2010bc09e7244a1ef67d173db31e70b20512e29757b91b614a88f": {
    "Name": "devops-app",
    "EndpointID": "82855ea7d7cea154d92aa285ccc9881690e258a39436be4ba81dbdd16a706f6c",
    "MacAddress": "a6:4a:c4:8f:45:6f",
    "IPv4Address": "172.21.0.3/16",
    "IPv6Address": ""
  }
},
"Status": {
  "IPAM": {
    "Subnets": {
      "172.21.0.0/16": {
        "IPsInUse": 5,
        "DynamicIPsAvailable": 65531
      }
    }
  }
}
```

Activate Windows
Go to Settings to activate Windows

```
"Status": {
  "IPAM": {
    "Subnets": {
      "172.21.0.0/16": {
        "IPsInUse": 5,
        "DynamicIPsAvailable": 65531
      }
    }
  }
}
}
}
}
]
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker volume ls
DRIVER      VOLUME NAME
local       cleanup-volume-1
local       cleanup-volume-2
local       cleanup-volume-3
local       day9-postgres-data
local       devops-db-data
local       devops-docker-final-project_postgres-data
local       docker-compose-day8_devops-db-data
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose ps
NAME                IMAGE              COMMAND                SERVICE    CREATED        STATUS              PORTS
devops-app           devops-final:v1    "python app.py"        app        6 minutes ago  Up 6 minutes        0.0.
0.0:5000->5000/tcp, [::]:5000->5000/tcp
devops-database      postgres:16-alpine "docker-entrypoint.s..." database    29 minutes ago  Up 29 minutes      0.0.
0.0:5432->5432/tcp, [::]:5432->5432/tcp
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$
```

Activate Windows
Go to Settings to activate Windows

TASK 8 — Failure Simulation

```

akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose up -d
[+] up 2/2
  Container devops-database Healthy 11.2s
  Container devops-app Started 11.8s
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose ps
NAME                IMAGE                COMMAND                SERVICE    CREATED        STATUS                PORT
devops-app          devops-final:v1      "python app.py"        app        36 seconds ago  Up 24 seconds        0.0.
0.0:5000->5000/tcp, [::]:5000->5000/tcp
devops-database     postgres:16-alpine   "docker-entrypoint.s..." database    23 minutes ago  Up 23 minutes (healthy) 0.0.
0.0:5432->5432/tcp, [::]:5432->5432/tcp
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker ps
CONTAINER ID        IMAGE                COMMAND                CREATED        STATUS                PORTS
f98f68809be2      devops-final:v1      "python app.py"        55 seconds ago  Up 43 seconds        0.0.0.0:5000->
>5000/tcp, [::]:5000->5000/tcp
04a06a87e853      postgres:16-alpine   "docker-entrypoint.s..." 23 minutes ago  Up 23 minutes (healthy) 0.0.0.0:5432->
>5432/tcp, [::]:5432->5432/tcp
73ecb58f639e      docker-compose-day9-web "/docker-entrypoint..." 23 hours ago    Up 3 hours (healthy) 0.0.0.0:9090->
>80/tcp, [::]:9090->80/tcp
f5953c4125a7      postgres:16-alpine   "docker-entrypoint.s..." 23 hours ago    Up 3 hours (healthy) 5432/tcp
day9-db
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose logs app
devops-app | * Serving Flask app 'app'
devops-app | * Debug mode: off
devops-app | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.

```

```

devops-app | * Running on all addresses (0.0.0.0)
devops-app | * Running on http://127.0.0.1:5000
devops-app | * Running on http://172.21.0.3:5000
devops-app | Press CTRL+C to quit
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose logs -f app
devops-app | * Serving Flask app 'app'
devops-app | * Debug mode: off
devops-app | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
devops-app | * Running on all addresses (0.0.0.0)
devops-app | * Running on http://127.0.0.1:5000
devops-app | * Running on http://172.21.0.3:5000
devops-app | Press CTRL+C to quit
^C
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose logs database
devops-database | The files belonging to this database system will be owned by user "postgres".
devops-database | This user must also own the server process.
devops-database |
devops-database | The database cluster will be initialized with locale "en_US.utf8".
devops-database | The default database encoding has accordingly been set to "UTF8".
devops-database | The default text search configuration will be set to "english".
devops-database |
devops-database | Data page checksums are disabled.
devops-database |
devops-database | fixing permissions on existing directory /var/lib/postgresql/data ... ok
devops-database | creating subdirectories ... ok
devops-database | selecting dynamic shared memory implementation ... posix
devops-database | selecting default max_connections ... 100

```

```

devops-database | 2026-08-19 08:56:52.557 UTC [1] LOG: listening on IPv4 address "0.0.0.0", port 5432
devops-database | 2026-08-19 08:56:52.557 UTC [1] LOG: listening on IPv6 address ":::", port 5432
devops-database | 2026-08-19 08:56:52.576 UTC [1] LOG: listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
devops-database | 2026-08-19 08:56:52.592 UTC [57] LOG: database system was shut down at 2026-08-19 08:56:52 UTC
devops-database | 2026-08-19 08:56:52.602 UTC [1] LOG: database system is ready to accept connections
devops-database | 2026-08-19 09:02:17.523 UTC [55] LOG: checkpoint starting: time
devops-database | 2026-08-19 09:02:20.524 UTC [55] LOG: checkpoint complete: wrote 31 buffers (0.2%); 0 WAL file(s) added, 0 removed, 0 recycled; write=2.816 s, sync=0.087 s, total=3.001 s; sync files=11, longest=0.021 s, average=0.008 s; distance=139 kB, estimate=139 kB; lsn=0/1948D90, redo lsn=0/1948D58
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ curl -i http://localhost:5000/health
HTTP/1.1 200 OK
Server: Werkzeug/3.1.8 Python/3.12.14
Date: Wed, 19 Aug 2026 09:24:17 GMT
Content-Type: application/json
Content-Length: 45
Connection: close

{"application": "running", "status": "healthy"}
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose ps

```

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS	PORT
devops-app	devops-final:v1	"python app.py"	app	5 minutes ago	Up 4 minutes	0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
devops-database	postgres:16-alpine	"docker-entrypoint.s..."	database	27 minutes ago	Up 27 minutes (healthy)	0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp

```

akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker inspect --format='{{.State.Health.Status}}' devops-database
healthy
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose exec database pg_isready -U devops -d devopsdb
/var/run/postgresql:5432 - accepting connections
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker network ls

```

NETWORK ID	NAME	DRIVER	SCOPE
eb0a7452964a	bridge	bridge	local
94b570376518	devops-docker-final-project_devops-network	bridge	local
5777044da3bd	docker-compose-day8_default	bridge	local
7aa49dac030d	docker-compose-day9_app-network	bridge	local
4129d5dc9746	host	host	local
41c3124c4dc7	none	null	local

```

akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker network inspect devops-docker-final-project_devops-network
[
  {
    "Name": "devops-docker-final-project_devops-network",
    "Id": "94b570376518f9f7dcf8dce757769f312b0a738e9db6b47833c6f9aeed8d7260",
    "Created": "2026-08-19T01:56:43.651133876-07:00",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [

```

```

akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker volume ls
DRIVER      VOLUME NAME
local       cleanup-volume-1
local       cleanup-volume-2
local       cleanup-volume-3
local       day9-postgres-data
local       devops-db-data
local       devops-docker-final-project_postgres-data
local       docker-compose-day8_devops-db-data
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose ps
NAME                IMAGE              COMMAND                SERVICE    CREATED          STATUS              PORT
devops-app           devops-final:v1    "python app.py"        app        6 minutes ago    Up 6 minutes        0.0.
0.0:5000->5000/tcp, [::]:5000->5000/tcp
devops-database      postgres:16-alpine "docker-entrypoint.s..." database    29 minutes ago    Up 29 minutes (healthy) 0.0.
0.0:5432->5432/tcp, [::]:5432->5432/tcp
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ nano docker-compose.yml
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose ps
NAME                IMAGE              COMMAND                SERVICE    CREATED          STATUS              PORT
devops-app           devops-final:v1    "python app.py"        app        17 minutes ago    Up 17 minutes        0.0.
0.0:5000->5000/tcp, [::]:5000->5000/tcp
devops-database      postgres:16-alpine "docker-entrypoint.s..." database    39 minutes ago    Up 39 minutes (healthy) 0.0.
0.0:5432->5432/tcp, [::]:5432->5432/tcp
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ curl http://localhost:5000/health
{"application":"running","status":"healthy"}
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ nano docker-compose.yml
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ nano docker-compose.yml

```

```

akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ nano docker-compose.yml
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose down
[+] down 3/3
  Container devops-app           Removed      10.6s
  Container devops-database      Removed      0.7s
  Network devops-docker-final-project_devops-network Removed      0.2s
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose up -d
[+] up 3/3
  Network devops-docker-final-project_devops-network Created      0.2s
  Container devops-database      Healthy       11.6s
  Container devops-app           Started       11.7s
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose ps
NAME                IMAGE              COMMAND                SERVICE    CREATED          STATUS              PORT
devops-app           devops-final:v1    "python app.py"        app        23 seconds ago    Up 11 seconds        0.0.
0.0:5000->5000/tcp, [::]:5000->5000/tcp
devops-database      postgres:16-alpine "docker-entrypoint.s..." database    23 seconds ago    Up 22 seconds (healthy) 0.0.
0.0:5432->5432/tcp, [::]:5432->5432/tcp
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose logs app
devops-app | * Serving Flask app 'app'
devops-app | * Debug mode: off
devops-app | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
devops-app | * Running on all addresses (0.0.0.0)
devops-app | * Running on http://127.0.0.1:5000
devops-app | * Running on http://172.21.0.3:5000
devops-app | Press CTRL+C to quit

```

```

akhil_chary@DESKTOP-1EMNQE8:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose logs database
devops-database | PostgreSQL Database directory appears to contain a database; Skipping initialization
devops-database |
devops-database | 2026-08-19 10:38:34.797 UTC [1] LOG: starting PostgreSQL 16.15 on x86_64-pc-linux-musl, compiled by gcc (Alpine 15.2.0) 15.2.0, 64-bit
devops-database | 2026-08-19 10:38:34.798 UTC [1] LOG: listening on IPv4 address "0.0.0.0", port 5432
devops-database | 2026-08-19 10:38:34.798 UTC [1] LOG: listening on IPv6 address "::", port 5432
devops-database | 2026-08-19 10:38:34.805 UTC [1] LOG: listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
devops-database | 2026-08-19 10:38:34.820 UTC [29] LOG: database system was shut down at 2026-08-19 10:38:18 UTC
devops-database | 2026-08-19 10:38:34.836 UTC [1] LOG: database system is ready to accept connections
akhil_chary@DESKTOP-1EMNQE8:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose config
name: devops-docker-final-project
services:
  app:
    build:
      context: /mnt/c/WINDOWS/system32/devops-docker-final-project
      dockerfile: Dockerfile
    container_name: devops-app
    depends_on:
      database:
        condition: service_healthy
        required: true
    environment:
      APP_ENV: development
      DATABASE_HOST: database
      DATABASE_NAME: devopsdb
      DATABASE_PASSWORD: devops_password

```

Activate Windows
Go to Settings to activate Windows.

```

  DATABASE_PORT: "5432"
  DATABASE_USER: devops
image: devops-final:v1
networks:
  devops-network: null
ports:
  - mode: ingress
    target: 5000
    published: "5000"
    protocol: tcp
restart: unless-stopped
database:
  container_name: devops-database
  environment:
    POSTGRES_DB: devopsdb
    POSTGRES_PASSWORD: wrong_password
    POSTGRES_USER: devops
  healthcheck:
    test:
      - CMD-SHELL
      - pg_isready -U devops -d devopsdb
    timeout: 5s
    interval: 10s
    retries: 5
image: postgres:16-alpine
networks:
  devops-network: null
ports:
  - mode: ingress

```

```

    name: devops-docker-final-project_devops-network
    driver: bridge
volumes:
  postgres-data:
    name: devops-docker-final-project_postgres-data
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose down -v
[+] down 4/4
  [x] Container devops-app Removed 10.5s
  [x] Container devops-database Removed 0.4s
  [x] Network devops-docker-final-project_devops-network Removed 0.2s
  [x] Volume devops-docker-final-project_postgres-data Removed 0.1s
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose up -d
[+] up 4/4
  [x] Network devops-docker-final-project_devops-network Created 0.1s
  [x] Volume devops-docker-final-project_postgres-data Created 0.0s
  [x] Container devops-database Healthy 11.6s
  [x] Container devops-app Started 12.4s
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose ps
NAME                IMAGE                                COMMAND                                SERVICE    CREATED          STATUS              PORT
devops-app          devops-final:v1                    "python app.py"                      app        23 seconds ago   Up 11 seconds       0.0.
0.0:5000->5000/tcp, [::]:5000->5000/tcp
devops-database     postgres:16-alpine                "docker-entrypoint.s..."           database   23 seconds ago   Up 22 seconds (healthy) 0.0.
0.0:5432->5432/tcp, [::]:5432->5432/tcp
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose logs database
devops-database    | The files belonging to this database system will be owned by user "postgres".
devops-database    | This user must also own the server process.
devops-database    |
devops-database    | The database cluster will be initialized with locale "en_US.utf8".

```

```

devops-database    | server stopped
devops-database    |
devops-database    | PostgreSQL init process complete; ready for start up.
devops-database    |
devops-database    | 2026-08-19 10:41:01.937 UTC [1] LOG:  starting PostgreSQL 16.15 on x86_64-pc-linux-musl, compiled by gcc
c (Alpine 15.2.0) 15.2.0, 64-bit
devops-database    | 2026-08-19 10:41:01.937 UTC [1] LOG:  listening on IPv4 address "0.0.0.0", port 5432
devops-database    | 2026-08-19 10:41:01.937 UTC [1] LOG:  listening on IPv6 address ":::", port 5432
devops-database    | 2026-08-19 10:41:01.946 UTC [1] LOG:  listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
devops-database    | 2026-08-19 10:41:01.958 UTC [57] LOG:  database system was shut down at 2026-08-19 10:41:01 UTC
devops-database    | 2026-08-19 10:41:01.974 UTC [1] LOG:  database system is ready to accept connections
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose logs app
devops-app         | * Serving Flask app 'app'
devops-app         | * Debug mode: off
devops-app         | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
devops-app         | * Running on all addresses (0.0.0.0)
devops-app         | * Running on http://127.0.0.1:5000
devops-app         | * Running on http://172.21.0.3:5000
devops-app         | Press CTRL+C to quit
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ curl http://localhost:5000/
<h1>DevOps Training Application</h1>
<p><strong>Application Status:</strong> Running</p>
<p><strong>Environment:</strong> Development</p>
<p><strong>Database:</strong> Connected</p>
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ curl http://localhost:5000/health
{"application":"running","status":"healthy"}
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ nano docker-compose.yml

```

```

akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose down -v
[+] down 4/4
  Container devops-app                        Removed      10.4s
  Container devops-database                  Removed      0.5s
  Volume devops-docker-final-project_postgres-data Removed      0.1s
  Network devops-docker-final-project_devops-network Removed      0.2s
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose up -d
[+] up 4/4
  Network devops-docker-final-project_devops-network Created      0.2s
  Volume devops-docker-final-project_postgres-data Created      0.0s
  Container devops-database                  Healthy      11.4s
  Container devops-app                      Started      11.6s
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose ps
NAME                IMAGE                COMMAND                SERVICE    CREATED        STATUS        PORT
devops-app          devops-final:v1      "python app.py"        app        22 seconds ago Up 9 seconds   0.0.
devops-database     postgres:16-alpine   "docker-entrypoint.s..." database    22 seconds ago Up 21 seconds (healthy) 0.0.
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose logs database
devops-database    | The files belonging to this database system will be owned by user "postgres".
devops-database    | This user must also own the server process.
devops-database    |
devops-database    | The database cluster will be initialized with locale "en_US.utf8".
devops-database    | The default database encoding has accordingly been set to "UTF8".
devops-database    | The default text search configuration will be set to "english".
devops-database    |
devops-database    | Data page checksums are disabled.
devops-database    |
                                                                Activate Windows
                                                                Go to Settings to activate Windows.

devops-database    | 2026-08-19 10:44:54.603 UTC [41] LOG:  database system is shut down
devops-database    | done
devops-database    | server stopped
devops-database    |
devops-database    | PostgreSQL init process complete; ready for start up.
devops-database    |
devops-database    | 2026-08-19 10:44:54.709 UTC [1] LOG:  starting PostgreSQL 16.15 on x86_64-pc-linux-musl, compiled by gcc (Alpine 15.2.0) 15.2.0, 64-bit
devops-database    | 2026-08-19 10:44:54.710 UTC [1] LOG:  listening on IPv4 address "0.0.0.0", port 5432
devops-database    | 2026-08-19 10:44:54.710 UTC [1] LOG:  listening on IPv6 address ":::", port 5432
devops-database    | 2026-08-19 10:44:54.717 UTC [1] LOG:  listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
devops-database    | 2026-08-19 10:44:54.727 UTC [57] LOG:  database system was shut down at 2026-08-19 10:44:54 UTC
devops-database    | 2026-08-19 10:44:54.740 UTC [1] LOG:  database system is ready to accept connections
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose logs app
devops-app          | * Serving Flask app 'app'
devops-app          | * Debug mode: off
devops-app          | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
devops-app          | * Running on all addresses (0.0.0.0)
devops-app          | * Running on http://127.0.0.1:5000
devops-app          | * Running on http://172.21.0.3:5000
devops-app          | Press CTRL+C to quit
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ curl http://localhost:8080/
<h1>DevOps Training Application</h1>
<p><strong>Application Status:</strong> Running</p>
<p><strong>Environment:</strong> Development</p>
<p><strong>Database:</strong> Connected</p>
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$

```

TASK 9 — Test Persistence

```

akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose ps
NAME                IMAGE                                COMMAND                                SERVICE  CREATED      STATUS      PORT
devops-database     2026-08-19 10:44:54.717 UTC [1] LOG: listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
devops-app          devops-final:v1                    "python app.py"                      app      39 minutes ago Up 39 minutes 0.0.
0.0:8080->5000/tcp, [::]:8080->5000/tcp740 UTC [1] LOG: database system is ready to accept connections
devops-database     postgres:16-alpine                "docker-entrypoint.s..."           database 39 minutes ago Up 39 minutes (healthy) 0.0.
0.0:5432->5432/tcp, [::]:5432->5432/tcp'
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose exec devops-database psql
-U devopsuser -d devopsdb
service "devops-database" is not running
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker volume ls
DRIVER      VOLUME NAME
local       cleanup-volume-1n http://172.21.0.3:5000
local       cleanup-volume-2 to quit
local       cleanup-volume-3B:/mnt/c/WINDOWS/system32/devops-docker-final-project$ curl http://localhost:8080/
local       day9-postgres-data
local       devops-db-dataApplication</h1>
local       devops-docker-final-project_postgres-data</p>
local       docker-compose-day8_devops-db-datament</p>
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose exec database \
> psql -U devopsuser -d devopsdb^Csystem32/devops-docker-final-project$ docker compose ps
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose exec database \
> psql -U devopsuser -d devopsdb
psql: error: connection to server on socket "/var/run/postgresql/.s.PGSQL.5432" failed: FATAL: role "devopsuser" does not
exist80->5000/tcp, [::]:8080->5000/tcp
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ nano docker-compose.yml

```

```

akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose exec devops-database psql
-U devopsuser -d devopsdb
service "devops-database" is not running
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker volume ls
DRIVER      VOLUME NAME
local       cleanup-volume-1n http://172.21.0.3:5000
local       cleanup-volume-2 to quit
local       cleanup-volume-3B:/mnt/c/WINDOWS/system32/devops-docker-final-project$ curl http://localhost:8080/
local       day9-postgres-data
local       devops-db-dataApplication</h1>
local       devops-docker-final-project_postgres-data</p>
local       docker-compose-day8_devops-db-datament</p>
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose exec database \
> psql -U devopsuser -d devopsdb^Csystem32/devops-docker-final-project$ docker compose ps
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose exec database \
> psql -U devopsuser -d devopsdb
psql: error: connection to server on socket "/var/run/postgresql/.s.PGSQL.5432" failed: FATAL: role "devopsuser" does not
exist80->5000/tcp, [::]:8080->5000/tcp
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ nano docker-compose.yml
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose exec database \
sql -U > psql -U devopsuser -d devopsdb^Csystem32/devops-docker-final-project$ docker compose up -d

```

```

akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose exec database \
sql -U > psql -U devops -d devopsdb
psql (16.15)
Type "help" for help.

devopsdb=# CREATE TABLE persistence_test (
devopsdb(#      id SERIAL PRIMARY KEY,
devopsdb(#      message VARCHAR(255)
devopsdb(# );
CREATE TABLE
devopsdb=# INSERT INTO persistence_test(message)
devopsdb=# VALUES ('This data must survive container recreation');
INSERT 0 1
devopsdb=# SELECT * FROM persistence_test;
 id |
-----+-----
  1 | This data must survive container recreation
(1 row)

devopsdb=# \q
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose down
[+] down 3/3
  Container devops-app      Removed      10.7s
  Container devops-database Removed      0.6s
  Network devops-docker-final-project_devops-network Removed      0.2s

```

```

akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose down
[+] down 3/3
  Container devops-app                      Removed          10.7s
  Container devops-database                 Removed          0.6s
  Network devops-docker-final-project_devops-network Removed          0.2s
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose up -d
[+] up 3/3
  Network devops-docker-final-project_devops-network Created          0.1s
  Container devops-database                 Healthy          11.4s
  Container devops-app                      Started          11.9s
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose ps
NAME                IMAGE                COMMAND                SERVICE    CREATED        STATUS                PORT
devops-app          devops-final:v1      "python app.py"        app        15 seconds ago Up 4 seconds          0.0.
devops-database     postgres:16-alpine   "docker-entrypoint.s..." database    15 seconds ago Up 15 seconds (healthy) 0.0.

```

```

akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ ocker compose exec database \
> psql -U devops -d devopsdb \
> ^C
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ docker compose exec database \
> psql -U devops -d devopsdb \
> -c "SELECT * FROM persistence_test;"
 id |
-----+-----
  1 | This data must survive container recreation
(1 row)

```

TASK 10 — Git Workflow

```

akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   Dockerfile
    modified:   docker-compose.yml

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    app/app.py
    app/requirements.txt

no changes added to commit (use "git add" and/or "git commit -a")
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git remote -v
origin  https://github.com/akhilchary81-lab/devops-docker-final-project.git (fetch)
origin  https://github.com/akhilchary81-lab/devops-docker-final-project.git (push)
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git branch
* main
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git pull origin main
From https://github.com/akhilchary81-lab/devops-docker-final-project
 * branch            main       -> FETCH_HEAD
Already up to date.
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git checkout -b develop
Switched to a new branch 'develop'
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git push -u origin develop
Username for 'https://github.com': akhilchary81-lab

```

```

akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git branch
* develop
  main
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git checkout develop
M       Dockerfile
M       docker-compose.yml
Already on 'develop'
Your branch is up to date with 'origin/develop'.
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git checkout -b feature/docker-app
Switched to a new branch 'feature/docker-app'
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git branch
  develop
* feature/docker-app
  main
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ ls
Dockerfile  README.md  app  docker-compose.yml  docs  scripts

```

```

akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git status
On branch feature/docker-app
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   Dockerfile
        modified:   docker-compose.yml

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        app/app.py
        app/requirements.txt

no changes added to commit (use "git add" and/or "git commit -a")
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git add .
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git commit -m "feat: add application"
[feature/docker-app d355a97] feat: add application
 4 files changed, 96 insertions(+)
 create mode 100644 app/app.py
 create mode 100644 app/requirements.txt
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git push -u origin feature/docker-app
Username for 'https://github.com': akhilchary81-lab
Password for 'https://akhilchary81-lab@github.com':
Enumerating objects: 11, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 4 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (7/7), 1.38 KiB | 37.00 KiB/s, done.
Total 7 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'feature/docker-app' on GitHub by visiting:

```

Activate Windows
Go to Settings to activate Windows

```

To https://github.com/akhilchary81-lab/devops-docker-final-project.git
* [new branch]      feature/docker-app -> feature/docker-app
branch 'feature/docker-app' set up to track 'origin/feature/docker-app'.
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git checkout develop
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git checkout -b feature/database
Switched to a new branch 'feature/database'
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git status
On branch feature/database
nothing to commit, working tree clean
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git add .
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git commit -m "feat: add database service"
On branch feature/database
nothing to commit, working tree clean
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git push -u origin feature/database
Username for 'https://github.com': akhilchary81-lab
Password for 'https://akhilchary81-lab@github.com':
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'feature/database' on GitHub by visiting:
remote:   https://github.com/akhilchary81-lab/devops-docker-final-project/pull/new/feature/database
remote:
To https://github.com/akhilchary81-lab/devops-docker-final-project.git
* [new branch]      feature/database -> feature/database
branch 'feature/database' set up to track 'origin/feature/database'.
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git status
On branch feature/database
Your branch is up to date with 'origin/feature/database'.
nothing to commit, working tree clean

```

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```

akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git add docker-compose.yml
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git commit -m "feat: add persistent volume"
On branch feature/database
Your branch is up to date with 'origin/feature/database'.

nothing to commit, working tree clean
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git push
Username for 'https://github.com': akhilchary81-lab
Password for 'https://akhilchary81-lab@github.com':
Everything up-to-date
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git checkout develop
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git checkout -b bugfix/health-check
Switched to a new branch 'bugfix/health-check'
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git status
On branch bugfix/health-check
nothing to commit, working tree clean
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git add .
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git commit -m "feat: add application health check"
On branch bugfix/health-check
nothing to commit, working tree clean
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git push -u origin bugfix/health-check
Username for 'https://github.com': akhilchary81-lab
Password for 'https://akhilchary81-lab@github.com':
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'bugfix/health-check' on GitHub by visiting:
remote:   https://github.com/akhilchary81-lab/devops-docker-final-project/pull/new/bugfix/health-check
remote:
To https://github.com/akhilchary81-lab/devops-docker-final-project.git
 * [new branch]      bugfix/health-check -> bugfix/health-check

```

Activate Windows
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```

akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git branch
* bugfix/health-check
  develop
  feature/database
  feature/docker-app
  main
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git branch -r
origin/HEAD -> origin/main
origin/bugfix/health-check
origin/develop
origin/feature/database
origin/feature/docker-app
origin/main
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ git log --oneline --all --decorate --graph
* d355a97 (origin/feature/docker-app, feature/docker-app) feat: add application
* ab1045f (HEAD -> bugfix/health-check, origin/main, origin/feature/database, origin/develop, origin/bugfix/health-check, origin/HEAD, main, feature/database, develop) Add environment configuration template
* 929300c Add project directories
* 803f1f3 Initialize Docker final project structure
* f101ee4 Initial commit
akhil_chary@DESKTOP-1EMNQEB:/mnt/c/WINDOWS/system32/devops-docker-final-project$ ^C

```

TASK 11 — Final Docker Troubleshooting Test

Root Cause:

The application could not connect to PostgreSQL because the configured database hostname did not match the Docker Compose service name.

Solution:

Updated the application database hostname to the correct Compose service name and recreated the application container.

Verification:

docker compose ps showed both services running.

docker compose logs app showed no connection errors.

The application responded successfully through the health endpoint.

Database connectivity was also verified successfully.

Problem

↓

docker ps

↓

docker ps -a

↓

docker compose ps

↓

docker logs

↓

docker compose logs

↓

docker inspect

↓

docker compose config

↓

docker network ls

↓

docker network inspect

↓

docker volume ls

↓

Identify Root Cause

↓

Apply Solution

↓

docker compose up -d --build

↓

Check containers

↓

Check logs

↓

Test application

↓

Test database

↓

Document

↓

Root Cause + Solution + Verification